Zhenyu You

Schiller Institute for Integrated Science and Society
Department of Earth and Environmental Sciences
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EDUCATION

Ph.D., Earth and Atmospheric Sciences 2020 – 2025
Georgia Institute of Technology, Atlanta, GA, USA
"Dynamical Origins of Warm-Season Precipitation Extremes over the Northern Extratropics: A Multiscale Diagnostic and Modeling Study"
Advisor: Dr. Yi Deng

M.S., Statistics 2022 – 2024
Georgia Institute of Technology, Atlanta, GA, USA

B.S., Atmospheric Sciences 2016 – 2020
Nanjing University, Nanjing, Jiangsu, China

EMPLOYMENT

Postdoctoral Research Fellow 09/2025 – present
Boston College, Chestnut Hill, MA, USA
Supervisor: Dr. Yi Ming

Graduate Research Assistant 08/2021 – 07/2025
Georgia Institute of Technology, Atlanta, GA, USA
Advisor: Dr. Yi Deng

PUBLICATIONS

You, Z., Lin, P., Peatman, S., and Ming, Y. (2026). Large-Scale Modulation of Mesoscale Convective System Propagation and Diurnal Rainfall over the Western Maritime Continent. Preprint, *EarthArxiv*.

Hwang, J., Deng, Y., Yook, S., and **You, Z.** (2026). Subseasonal prediction of the wintertime North Pacific blocking in AI-based weather models. Journal of Geophysical Research: Atmospheres, 131 (15), e2026JD046695.

You, Z., Deng, Y., Ming, Y., and Hwang, J. (2025). More Frequent Extreme Precipitation on the Asian Monsoon Fringes Driven by Evolving Extratropical Planetary-Scale Circulations. Preprint, *EarthArXiv*.

Hwang, J., **You, Z.,** Deng, Y., and Kim, H. (2025). Unfolding North American Spring Weather Extremes along a Scale Ladder. *Scientific Reports*, 15, 38729.

Hwang, J., **You, Z.**, He, J., and Deng, Y. (2025). Dynamics and Model Representation of the Boreal Spring Subseasonal Variability over North America. *Climate Dynamics*, 63, 360.

Park, C., Ming, Y., Dong, W., Deng, Y., and **You, Z.** (2025). Precipitation and Lifecycle Characteristics of US Mesoscale Convective Systems Simulated in a High-Resolution GFDL AM4. *Journal of Climate*, 38, 5901-5922.

You, Z., and Deng, Y. (2025). A Linear Analysis of the Heating Footprints of the U.S. Mesoscale Convective Systems in the Northern Extratropical Synoptic Variability. *Journal of Climate*, 38, 3845–3859.

You, Z., Deng, Y., Ming, Y., and Dong, W. (2024). A Multiscale Assessment of the Springtime U.S. Mesoscale Convective Systems in the NOAA GFDL AM4. *Climate Dynamics*, 62, 4017–4030.

You, Z., and Deng, Y. (2023). A Hierarchical Dissection of Multiscale Forcing on the Springtime Mesoscale Convective Systems in the United States. *Journal of Climate*, 36, 39–54.

Cao, D., Park, T.W., Song, Y., Wang, Y., **You, Z.,** et al. (2021). A Dynamical Pathway Bridging African Biomass Burning and Asian Summer Monsoon. *Climate Dynamics*, 57, 1993–2004.

CONFERENCE PRESENTATIONS

Deng, Y. and **You Z.** (2026). Multi-scale Forcing and Convective Heating Feedback of the Springtime Mesoscale Convective Systems in the United States. *AOGS2026 (Oral).*

You, Z., Deng, Y., and Ming, Y. (2025). Changing Rossby Wave Pathways Behind the Increasing Frequency of Summertime Extreme Precipitation in Northeast China and Pakistan. *AGU Fall Meeting 2025 (Oral)*.

You, Z., and Deng, Y. (2025). A Linear Analysis of the Footprints of Heating from U.S. Mesoscale Convective Systems on Springtime Extratropical Synoptic Variability. *AMS Annual Meeting 2025 (Oral)*.

You, Z., Deng, Y., Ming, Y., and Dong, W. (2023). A Multiscale Assessment of the Springtime Mesoscale Convective Systems in the United States. *AGU Fall Meeting 2023 (Oral)*.

RESEARCH EXPERIENCE

Postdoctoral Research Scientist, Boston College 2025 – current

- Characterization of Mesoscale Convective Systems propagation and their impact on diurnal cycle
- Evaluation of Mesoscale Convective System simulation in high-resolution weather and climate models
- Diagnosis of the propagation mechanism of Mesoscale Convective Systems and squall lines.

Graduate Research Assistant, Georgia Institute of Technology 2021 – 2025

- Multiscale Forcing Mechanisms of U.S. Mesoscale Convective Systems
- Evaluation of Mesoscale Convective System Simulation in Earth System Models
- Heating Footprints of Mesoscale Convective Systems in Extratropical Synoptic Variability
- Drivers of Extreme Summertime Precipitation in Central and East Asia

Visiting Student, Georgia Institute of Technology 07/2019 – 09/2019

- Dynamical Pathway Bridging African Biomass Burning and Asian Summer Monsoon

Undergraduate Research Assistant, Nanjing University 2017 – 2020

- The Impact of ENSO on Vapor Transport and Atmospheric Rivers over the North Pacific

TEACHING EXPERIENCE

Climate & Global Change, Georgia Institute of Technology 2025

- Delivered a guest lecture titled "*Extreme Weather and Mechanisms in the Midlatitudes*".
- Led student discussions translating peer-reviewed research into policy-relevant summaries.

Habitable Planet, Georgia Institute of Technology 2021 – 2024

PROFESSIONAL DEVELOPMENT

Tropical dynamics workshop (organizer), Boston College	04/2026
CESM Tutorial, NSF NCAR Mesa Laboratory	07/2025
MPAS-Atmosphere Tutorial	04/2025
RossbyPalooza Summer School, the University of Chicago	07/2024
Neuromatch Academy Deep Learning	07/2023

AWARDS & HONORS

CESM Tutorial Travel Support, NSF NCAR	07/2025
RossbyPalooza Workshop Travel Support, the University of Chicago	07/2024
Douglas Davis Fellowship Award, Georgia Institute of Technology	04/2023
Merit Student of the School of Atmospheric Sciences, Nanjing University	04/2018
Tung OOCL Scholarship, Tung Foundation & Orient Overseas Container Line Ltd.	12/2017
Excellent Student Leader, School of Atmospheric Sciences, Nanjing University	09/2017

PROFESSIONAL MEMBERSHIPS

American Geophysical Union (2023 – present)
American Meteorological Society (2024 – present)